

Figure 10.3 Oscillatory or vibratory motions

10.2 SIMPLE HARMONIC MOTION (SHM)

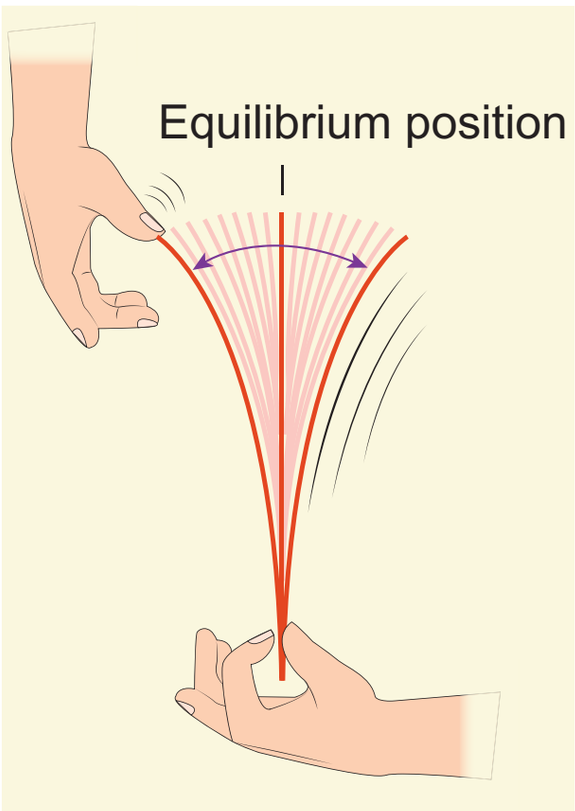


Figure 10.4 Simple Harmonic Motion

Note

A simple harmonic motion is a special type of oscillatory motion. But all oscillatory motions need not be simple harmonic .

Simple harmonic motion is a special type of oscillatory motion in which the acceleration or force on the particle is directly proportional to its displacement from a fixed point and is always directed towards that fixed point. In one dimensional case, let x be the displacement of the particle and a_x be the acceleration of the particle, then

$$a_x \propto x \qquad (10.1)$$

$$a_x = - \mathrm{b}\, x \qquad (10.2)$$

where b is a constant which measures acceleration per unit displacement and dimensionally it is equal to T^{-2}. By multiplying by mass of the particle on both sides of equation (10.2) and from Newton's second law, the force is

$$F_x = - k\, x \qquad (10.3)$$

where k is a force constant which is defined as force per unit length. The negative sign indicates that displacement and force (or acceleration) are in opposite directions. This means that when the displacement of the particle is taken towards right of equilibrium position (x takes positive value), the force (or acceleration) will point towards equilibrium (towards left) and similarly, when the

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